

Advanced Machine Learning Project Suggestions

Overview

Each team may select one of the following projects or propose their own topic, provided it is equally advanced and challenging. For each project below, we list representative medium-size datasets and two recent journal references to guide background reading.

1 Multi-Modal Fake News Detection

Objective: Build a classification system that leverages both textual and visual information to detect fake news, comparing early/late/cross-attention fusion against strong unimodal baselines.

Core Learning Outcomes:

- Multi-modal learning (text + image).
- Deep learning fine-tuning (Transformers, CNNs).
- Rigorous evaluation (ablation on fusion strategies).

Technical Approach (sketch):

1. Preprocess text (tokenization) and images (resize/normalize).
2. Extract features: pre-trained Transformer for text; pre-trained CNN for images.
3. Compare early fusion, late fusion, and cross-attention fusion.
4. Evaluate vs. unimodal baselines using Accuracy/F1.

Suggested medium-size datasets:

- **Fakeddit** (millions of multimodal posts; use a balanced medium subset).
- **Weibo** or **PHEME** (text + images; rumor/fake labels).

Recent references:

- Mousavi et al., E-CaTCH, <https://arxiv.org/pdf/2508.11197>
- Shen et al. (2024), *Frontiers in Computer Science*: contrastive/cross-modal fake news detection with optimal transport. <https://www.frontiersin.org/journals/computer-science/articles/10.3389/fcomp.2024.1473457/full>

2 Unsupervised Anomaly Detection in Financial Time-Series (News + Stocks)

Objective: Detect financial anomalies (e.g., volatility spikes) by integrating time-series market data with textual signals (news/tweets).

Core Learning Outcomes:

- Unsupervised anomaly detection in temporal data.
- Text–time-series feature fusion (e.g., sentiment + returns).
- Event-based evaluation.

Technical Approach (sketch):

1. Build features from prices/volumes and text (bag-of-words/sentiment).
2. Compare Isolation Forest, One-Class SVM, predictive LSTM/Transformer (flag anomalies via prediction error).
3. Evaluate with PR-AUC and event-based accuracy (e.g., high-volatility days).

Suggested medium-size datasets:

- **StockNet** (tweets + stock prices; medium-scale).
- **FiQA** (finance text) aligned with daily OHLCV series for S&P500 constituents.

3 Unsupervised Anomaly Detection in Healthcare Time-Series (ICU)

Objective: Benchmark unsupervised methods on ICU vital signs/labs; optionally fuse clinical notes to improve detection of deterioration.

Core Learning Outcomes:

- Temporal modeling of physiological signals.
- Multi-modal extension with clinical text.
- Case-based clinical evaluation.

Technical Approach (sketch):

1. Build time-series windows and optional text features (TF-IDF).
2. Compare autoencoder reconstruction vs. predictive LSTM/Transformer errors.
3. Evaluate with ROC-AUC/F1 and clinical case analyses (e.g., sepsis onset).

Suggested medium-size datasets:

- **MIMIC-III / MIMIC-IV** (ICU time-series + notes; large but subsettable to medium cohorts).
- **eICU-CRD** (multi-hospital ICU data with vitals/notes).

4 Dimensionality Reduction and Representation Learning

Objective: Compare PCA/ICA vs. nonlinear methods (t-SNE/UMAP) and assess impact on downstream classifiers.

Core Learning Outcomes:

- Latent representation analysis (interpretability vs. utility).
- Visualization in reduced spaces.
- Impact on learning efficiency/accuracy.

Technical Approach (sketch):

1. Compute 2D/3D embeddings (PCA/ICA/t-SNE/UMAP).
2. Train simple classifiers on reduced vs. original features.
3. Analyze trade-offs in accuracy, runtime, and interpretability.

Suggested medium-size datasets:

- **CIFAR-10** (50k images), **Fashion-MNIST** (60k), **20 Newsgroups** (~18k docs).

Recent references:

- Müller et al. (2023), *Analyst (RSC)*: comparative DR study in high-dimensional biomedical imaging.
- Rastogi et al. (2023), *Eng. Proc. (MDPI)*: overview of DR methods and practical issues.

5 Context-Aware Music Recommendation with Tensor Factorization

Objective: Go beyond matrix factorization by adding contextual dimensions (time-of-day, week-day/weekend, device), and optionally incorporate lyrics features.

Core Learning Outcomes:

- Tensor factorization (CP/Tucker) for recommendation.
- Context modeling and ranking evaluation (MAP@k, NDCG@k).
- Optional multi-modal extension via lyrics features.

Technical Approach (sketch):

1. Build a tensor $[\text{Users} \times \text{Songs} \times \text{Context}]$.
2. Baseline: matrix factorization on $[\text{Users} \times \text{Songs}]$.
3. Advanced: CP or Tucker decomposition; predict user-song-context relevance.
4. Extension: add lyrics features (bag-of-words/TF-IDF) as side information.

Suggested medium-size datasets:

- **Million Song Dataset (MSD)** features + **MusixMatch** lyrics (hundreds of thousands).
- **Last.fm 360k** (user-artist play counts + tags; alignable with lyrics).